Response to reviewers

Channel-Aware Backdoor Attacks Against Federated Infostealer Malware Classification Using Dynamic API-Call and Network Representations (IWBIS)

We thank all three reviewers. Every comment is answered in the table below, with the section of the revised paper that carries the change. Where we did not make a requested change, we say so and give the reason.

Table numbering has changed. The single-seed defense screening now precedes the multi-seed results, and the old Tables V and VII have been merged. Where a comment below cites a table by its old number, our reply gives the new one. Figure numbering is unchanged.

Submitted version	This version
Table I, dataset summary	Table I
Table II, experimental environment	Table II (condensed from seven rows to four)
Table III, backbone selection	Table III
Table IV, channel-aware backdoor sweep	Table IV (caption now states seed 42)
Table V, main multi-seed results	**Table VI**
Table VI, single-seed defense screening	**Table V**
Table VII, multi-seed Multi-Krum results	**merged into Table VI**

The merge removed only the two *Backdoor, FedAvg* rows that Table VII duplicated from Table V exactly. No result was dropped.

Comment	Response
Reviewer A	
Originality should be supported by a clear comparison with closely related recent studies.	Section II-C now positions the work against backdoor attacks on malware classifiers specifically: explanation-guided feature-space triggers (Severi et al., USENIX Security 2021) and family-selective triggers (Yang et al., IEEE S&P 2023). Both attack static features of a centralized model, whereas we attack a federated model through the channels of a dynamic-behavior image. The abstract now also states the novelty in one sentence.
The experimental procedure is not described with sufficient clarity; the stages are not explicitly defined.	Section III opens with a paragraph naming each block of Fig. 1 in the order the subsections take them. Section III-E gives the attack as implemented: 20% of the attacker's 14 AgentTesla training images, that is two images per round, resampled every round; a white 12×12 trigger written at the bottom-right corner after normalization; client 0 malicious in every round. Section III-F gives the defense parameters, including $f = 1$ and $\lvert\mathcal{S}\rvert = 2$ of five clients, and defines the trigger control. Section III-H states AdamW, ResNet18 from scratch, the final-round model with no best-validation selection, and that the per-round curves are test-set curves.

Comment	Response
The workflow figure is insufficiently integrated; it should be referenced and its blocks explained.	Fig. 1 is referenced in the first sentence of Section III, and the new opening paragraph walks its blocks in order.
Figure 2 is mentioned in II.B while it illustrates Dataset Creation, described in III.A.	Fixed. The reference to Fig. 2 now sits in Section III-A, next to the pipeline it depicts. Fig. 3 stays in III-B, where the representations are described.
Some tables are not mentioned anywhere in the paper.	All thirteen floats are now referenced at least once. Table I is cited in III-A, Table II in III-G and Fig. 4 in III-D.
The abstract should state the contribution or novelty more explicitly.	Added: the channels of the representation are treated as the attack surface, and a trigger confined to the fusion channel alone matches one spanning all three.
The discussion provides limited interpretation, and several claims lack supporting references.	Section IV now gives a measured mechanism for each main result. For the channel ordering, the mean intensity inside the trigger region over all 500 images is 241.7 in R, 81.5 in G and 10.2 in B, and 37.7% of red trigger-region pixels are already at or above 254. For Multi-Krum, the malicious client survives selection in 8% to 50% of rounds, and that rate correlates with the final ASR ($r = 0.893$, $p = 0.017$, $n = 6$), which we connect to [15] and [22].
The reference list should represent recent studies related to the method, and the discussion may need additional recent references.	We added the two malware-backdoor references above and now use [15] and [22] in the discussion. We removed two references doing no work: one cited only for the cross-entropy loss, and one background citation whose claim the two remaining citations already support. The list is 25 entries, the same count as before, with a more recent composition. We verified mechanically that every entry is cited and numbered in order of first citation.
Standard of English: Fair.	Two proofreading passes. Beyond ordinary corrections we unified the model notation, inconsistent between (1) and (3) to (5), fixed a citation used to support a claim it contradicts, and corrected a statement that images are normalized to $[0, 255]$ when the code normalizes them to $[-1, 1]$.
Reviewer B	
The author should conduct significance testing.	Section IV reports Fisher exact tests on the pooled triggered samples. Blue/fusion and full RGB differ from their trigger controls at $p = 3.5 \times 10^{-19}$ and $p = 2.6 \times 10^{-18}$, and blue differs from red and green at $p = 4.0 \times 10^{-8}$. Multi-Krum reduces ASR against FedAvg at $p = 1.5 \times 10^{-8}$ and $p = 9.1 \times 10^{-7}$. Pooling is legitimate because the split is re-drawn per seed, and the paper says so. Paired tests on clean performance are not significant, and are reported with the caveat that they are low-powered at $n = 3$. This changed one of our claims: red and green are indistinguishable from their own controls, both at $p = 1$, so the result is no detectable effect rather than a weaker one. Section IV-B now says that instead of "remain less effective".

Comment	Response
The author should explain the architecture of SmallCNN.	Section III-C describes it: three convolutional blocks of 32, 64 and 128 channels, each Conv 3×3 with batch normalization and ReLU, max-pooling after the first two, adaptive average pooling after the third, and a linear layer to five classes. All backbones are trained from scratch.
Tables I and II and Figures 1 and 4 exist but were never mentioned.	All four are now referenced, as are the remaining floats.
Keywords are too many.	Reduced from seven to five.
The author should mention and discuss every table and figure.	Every float is now referenced and discussed in at least one sentence.
Contribution and originality: the contributions are empirical experiments.	We accept that the contribution is empirical, and have made the claim sharper rather than broader: a trigger confined to the fusion channel alone is as effective as one spanning all three, even though that channel is derived from the other two and carries no independent information.
Reviewer C	
Backbone selection, the channel sweep and the single-seed defense screening all use seed 42, so the selection may be biased toward one seed.	The paper now states this rather than leaving it implicit. The screening has its own subsection (IV-C, Table V) ahead of the multi-seed results, with the reason given: screening every defense on every seed would multiply the training budget for candidates that may not survive, and all runs share one GPU. Section IV-E notes that the screening uses seed 42, precisely the seed at which blue/fusion with Multi-Krum fails. The captions of Tables III, IV and V all state seed 42.
Only three seeds; a deviation as large as 0.5111 ± 0.4286 suggests a bimodal outcome, not partial mitigation.	You are right, and this was the most useful correction we received. Table VI reports Multi-Krum per seed alongside the mean, with a note on both Multi-Krum rows stating that the across-seed ASR range is at least 0.5, so the mean describes no individual run. The per-seed values are 15/15, 3/15 and 5/15 for blue/fusion and 6/15, 15/15 and 6/15 for full RGB. Section IV-E and the conclusion now describe the outcome as bimodal, and the abstract no longer quotes the average ASR.
The attack tests only one source-target pair, so the generalization claim reaches beyond the evidence.	We have not added a second pair. The conclusion lists this as a limitation, and the abstract qualifies the claim as holding “in this controlled IID setting”.
The FL setting is IID-only although the motivation for FL is inherently non-IID.	Also not addressed by new experiments, and now stated as a limitation. The split files already carry a non-IID client assignment that these experiments do not use, so the experiment is available to future work rather than blocked.
No statistical significance testing is reported.	Added, as described under Reviewer B.
The attacker’s capability is underspecified; it is unclear whether model replacement or scaling as in [12] is employed.	Section III-E states that the capability is pure data poisoning: the malicious client trains and submits like any other, with no update scaling and no model replacement, so [12] is background rather than the attack implemented here.

Comment	Response
Multi-Krum on full RGB drops clean accuracy from 0.8400 to 0.7600, a trade-off underemphasized in the discussion and the abstract.	The drop is now in both. Section IV-E states it next to the ASR reduction, and the abstract reports it as part of the Multi-Krum result.
The conclusion states more confidence than $n = 3$ supports, and "partially suppresses" is imprecise.	The phrase is gone. The conclusion says that Multi-Krum reduced ASR bimodally rather than partially, leaving the backdoor fully effective at 15/15 on one of the three seeds for each trigger.
The conclusion does not address the alternative explanation that the blue channel's effectiveness may stem from its information content, which remains untested.	This turned out to be testable without new experiments, and the alternative can be ruled out. The dataset construction code shows the blue channel is the edge map of the average of the other two, $B = \mathrm{FindEdges}((R + G)/2)$, which we verified against the shipped dataset by recomputing it with zero difference on 400 of 400 images. Blue is therefore a deterministic function of red and green and carries no independent information, so its effectiveness cannot come from information the others lack. Section III-B gives the formula; Section IV-B draws the consequence, that what distinguishes blue is that it is almost empty, 54.5% of its pixels exactly zero. We state that this saliency explanation is consistent with the measurements but not established, and name the contrast-matched trigger as the experiment that would settle it.
The "serious threat" framing should be moderated.	The abstract and conclusion now say "a serious threat to federated malware classifiers in this controlled IID setting".
Changes no reviewer asked for	
The seed-42 clean baseline was trained on a different budget.	The clean FL baseline for seed 42, and the trigger controls evaluated against its checkpoint, come from a run of 30 rounds with one local epoch, while every other run uses 50 rounds with two. Section III-H now states this. Re-training under the common budget gives 0.7867 accuracy and 0.7869 macro-F1 with a control rate of 6/15 instead of 4/15, so it is worse than the run we report. We state both directions of the bias: the reported run sets a higher bar for the clean-performance claim and a lower one for the trigger-control argument. No ASR result is affected.
Table III was not a paired comparison.	The two representations were generated with different split files, so the rows were measured on different 75-image test sets and cannot support a performance claim across representations. Section III-C now says so, and the selection of RGB-stack rests on the design argument the paper already made, with macro-F1 selecting only the backbone within that representation. The abstract, Section IV-A and the conclusion were adjusted to match.

Comment	Response
Figure 7 was regenerated.	The submitted version was exported at 8.5 inches wide and then placed in a 3.5 inch column, so its tick labels printed at about 2.3 pt against 10 pt body text. It is now generated at the size it is placed at and prints at 6 to 7 pt. The data is unchanged: the final-round ASR of all four curves reproduces Table VI exactly.
What we did not change	
More seeds, a second source-target pair, and a non-IID partition.	These need experiments we did not run for this revision. All three are stated as limitations in the conclusion, together with the note that the red and green results rest on seed 42 alone, and the three matching experiments are named as future work. We would rather mark the boundary of the evidence than argue past it.